

11 — API Documentation & Endpoint Reference

■ REST API Architecture Overview

The backend service hosts fully documented REST API endpoints running on **CODE_PH_0**. Interactive Swagger UI documentation is available at **CODE_PH_1**.

■ Complete REST Endpoint Reference Table

Method	Endpoint	Purpose / Function	Input Parameters	Output Payload	Frontend Usage
CODE_PH_0	CODE_PH_0	Root status check	None	CODE_PH_0	Server verification
CODE_PH_0	CODE_PH_0	Liveness check	None	CODE_PH_0	Container health
CODE_PH_0	CODE_PH_0	List available commodities	None	CODE_PH_0	Tab 3 Dropdown
CODE_PH_0	CODE_PH_0	Data-grounded ML cargo forecast	CODE_PH_0 , CODE_PH_1 , CODE_PH_2	Forecast series, 95% bounds, drivers, NMPA meta	■ Tab 3 Main Chart
CODE_PH_0	CODE_PH_0	Backtesting WAPE comparison	CODE_PH_0 , CODE_PH_1	Test actuals, baseline vs ML model, metrics table	Tab 3 Backtesting
CODE_PH_0	CODE_PH_0	Driver weights & explanations	CODE_PH_0 , CODE_PH_1	Drivers list with weights and directions	Tab 3 Drivers
CODE_PH_0	CODE_PH_0	Lineage & completeness stats	None	Completeness %, record count, date span	Tab 3 Lineage Card
CODE_PH_0	CODE_PH_0	What-If cargo scenario simulator	CODE_PH_0 , CODE_PH_1 , CODE_PH_2	Simulated throughput, delta tonnes, stress level	Tab 3 Simulator
CODE_PH_0	CODE_PH_0	Berth occupancy status	None	List of 12 berths with commodity assignments	Tab 1 Berth Bar Chart
CODE_PH_0	CODE_PH_0	Active AIS vessel logs	None	List of 25 vessels with coords, ETA, delay prob	Tab 1 & Tab 2 Map/Table
CODE_PH_0	CODE_PH_0	Congestion risk warnings	None	List of alert objects with risk scores	Tab 2 Alert Cards

CODE_PH_0	CODE_PH_0	Trade lane volume & growth %	None	List of trade routes with growth rates	Tab 4 Trade Lanes
CODE_PH_0	CODE_PH_0	30-day commodity price trends	None	Price USD series, 30d delta	Tab 4 Price Ticker
CODE_PH_0	CODE_PH_0	AI-ranked trade opportunities	None	Ranked list with revenue potential in ■ Cr	Tab 4 Opportunities
CODE_PH_0	CODE_PH_0	Active anomaly event logs	None	Anomalies classified by severity (Surge, Decline)	Tab 5 Anomaly Cards
CODE_PH_0	CODE_PH_0	30-day anomaly histogram data	None	Historical count by severity tier	Tab 5 Histogram
CODE_PH_0	CODE_PH_0	Trade incentive policy recs	None	Priority-ranked policy recommendations	Tab 6 Incentives
CODE_PH_0	CODE_PH_0	Monte Carlo revenue simulator	CODE_PH_0, CODE_PH_1, CODE_PH_2	P10/P50/P90 quantiles & 1,000 sample points	Tab 6 Histogram
CODE_PH_0	CODE_PH_0	Digital twin scenario result	CODE_PH_0 (CODE_PH_1, CODE_PH_2, etc.)	Congestion index, utilization, radar metrics	Tab 7 Digital Twin
CODE_PH_0	CODE_PH_0	Simulated berth utilization	None	Baseline vs simulated berth utilization	Tab 7 Berth Chart
CODE_PH_0	CODE_PH_0	Suggested AI questions	None	List of sample question strings	Tab 8 Copilot Chips
CODE_PH_0	CODE_PH_0	Natural language copilot query	CODE_PH_0	3-tier dispatch trace, confidence, answer	Tab 8 AI Response
CODE_PH_0	CODE_PH_0	Ingestion pipeline health	None	Kafka msg/sec, Spark rows/sec, storage GB	Tab 9 System Metrics
CODE_PH_0	CODE_PH_0	Live ingestion event log	None	List of timestamped ingestion log rows	Tab 9 Event Log
CODE_PH_0	CODE_PH_0	Executive KPI summary	None	Berth util, throughput, revenue, congestion	Tab 1 Top Metrics
CODE_PH_0	CODE_PH_0	30-day revenue & throughput	None	Dates, revenue ■ Cr, throughput MT arrays	Tab 1 Trend Line

CODE_PH_0	CODE_PH_0	Port event stream	None	Recent arrival/dep arture/forecast events	Tab 1 Event Log
CODE_PH_0	CODE_PH_0	Multipart CSV file upload	CODE_PH_0 (Multipart Form)	File path, row count, column headers	Data Integration

■ Detailed Explanation of Primary Core Endpoints

1. CODE_PH_0

- Query Parameters:
- **CODE_PH_0**: Integer (3, 6, 12 months, default: 6).
- **CODE_PH_0**: String (**CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**, **CODE_PH_4**, etc.).
- **CODE_PH_0**: String (**CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**).
- Example JSON Response:

```
{ "summary": { "current_monthly_volume_tonnes": 625768.0, "expected_monthly_avg_tonnes": 710889.9,
"total_expected_horizon_tonnes": 4265339.5, "forecast_change_pct": 13.6, "trend_signal": "Increasing /
Bullish", "predictability_level": "MEDIUM", "model_wape_pct": 17.15, "model_accuracy_pct": 82.8 },
"chart": { "history_months": ["2024-01", "2024-02", "2024-03"], "history_values": [657884.0, 612400.0,
640100.0], "forecast_months": ["2026-08", "2026-09", "2026-10"], "forecast_values": [690000.0,
715000.0, 730000.0], "lower_bounds": [610000.0, 625000.0, 635000.0], "upper_bounds": [770000.0,
805000.0, 825000.0] }, "nmpa_facility": { "facility_name": "Mechanized Coal Handling Terminal",
"berths": "Berth 15 & 16", "hinterland_consumer": "UPCL (Udupi Power Corp Ltd / Adani Power)",
"max_draft_m": 14.0 } }
```

2. CODE_PH_0

- Example Request Body:

```
{ "commodity": "TOTAL COAL", "section": "ALL", "vessel_arrival_change_pct": 15.0,
"trade_demand_change_pct": 10.0, "weather_delay_days": 1.0, "horizon_months": 6 }
```

- Example JSON Response:

```
{ "simulation_summary": { "baseline_total_tonnes": 4265339.5, "simulated_total_tonnes": 4836894.5,
"volume_delta_tonnes": 571555.0, "volume_delta_pct": 13.4, "capacity_risk_level": "MODERATE_STRESS" },
"simulated_operational_advisories": [ "Vessel arrival increase of 15.0% requires preparing 2 additional
bulk unloading cranes." ] }
```